Population Parameters and Sample Statistics

Global Economics | Grade 11 | Term 2 | Lesson 1

Name _____ **Date** _____

C1. Distinguishes population parameters from sample statistics by interpreting and comparing the notation for population and sample means, standard deviations, and proportions in economic and financial contexts.

Worked example A market association's sales survey

The **population** is all 1,200 stalls licensed for June, the group the association wants to describe. The **sample** is the 60 stalls it randomly selects. Of those 60, 18 accept digital payment.

Measure	Population parameter	Sample statistic
Mean June sales revenue	μ	$\bar{x} = 4.8$ million COP
Standard deviation of June sales revenue	σ	$s = 1.2$ million COP
Share that accept digital payment	p	$\hat{p} = 18/60 = 0.30 = 30\%$

The parameters describe all 1,200 stalls and their values are unknown from this survey. Among the 60 selected stalls, sales average 4.8 million COP with a standard deviation of 1.2 million COP. To get the sample's digital-payment share, divide the 18 stalls that accept digital payment by the 60 selected.

Guided practice

1. **A supermarket's checkout survey**
   A manager studies all 1,500 Saturday purchases at one store. Random samples A and B each contain 50 receipts from those purchases. A loyalty card was used in 20 purchases in A and 25 in B.

   (a) Identify the population and what each sample contains.

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   (b) Label each statistic with its symbol and complete the table. Purchase values are in thousand COP. Show the proportions as percentages and give their difference in percentage points.

Measure and symbol	A	B	B minus A
Mean purchase value _____	64	61	
Standard deviation _____	18	20	
Loyalty-card share _____			

   (c) Use the table to describe how the three measures differ in the samples. In this survey, what do μ, σ, and p describe, and why do their values stay fixed across the two draws?

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Independent practice

2. **A credit union's loan review**

   On June 30, a credit union reviews its 800 active microenterprise working-capital loans. The lending committee then broadens the review to include active microenterprise equipment loans. Together, the two categories contain 1,000 active loans on that date. For each review the credit union randomly selects 50 loans from the relevant list.

Measure	Initial sample	Broader sample
Mean outstanding balance (million COP)	8.0	8.8
Standard deviation of outstanding balances (million COP)	2.4	2.9
Loans with an overdue installment on June 30	10	8

   (a) Identify the population and sample in each review. Calculate the change in population size.

   (b) Give the symbol and meaning of each population parameter in the two reviews. Pair each parameter with its sample statistic. Calculate both sample proportions and interpret the three statistics in each sample.

   (c) Calculate each difference (broader sample minus initial sample), with units. Express the difference in overdue shares in percentage points. Explain what the data let you conclude about changes in the population parameters.

Solutions

Sample (green)	Population (red)
Sample mean ($\bar{x}$)	Population mean (μ)
Sample standard deviation (s)	Population standard deviation (σ)
Sample proportion ($\hat{p}$)	Population proportion (p)
Sample (selected units)	Population (all units of interest)

1. **A supermarket's checkout survey**

   (a) The population is all 1,500 Saturday purchases at the store. Samples A and B are two draws of 50 Saturday purchases, represented by the receipts.

   (b) Each difference is B minus A. The mean and standard deviation are in thousand COP. The share difference is in percentage points (pp).

Statistic	A	B	B minus A
Mean $\bar{x}$	64	61	$61 - 64 = -3$
Standard deviation s	18	20	$20 - 18 = +2$
Loyalty-card share $\hat{p}$	$20/50 =$ 40%	$25/50 =$ 50%	$50 - 40 = +10$ pp

   (c) Sample B's mean purchase value is 3 thousand COP lower, its standard deviation is 2 thousand COP higher, and its loyalty-card share is 10 percentage points higher. Across all 1,500 purchases, μ is the mean purchase value, σ is the standard deviation of purchase values, and p is the loyalty-card share. Each parameter is fixed because both samples come from the same set of purchases. The parameter values themselves are unknown.

2. **A credit union's loan review**

   (a) On June 30 the initial population is all 800 active microenterprise working-capital loans. The broader population is all 1,000 active loans across both categories. Each sample consists of 50 loans drawn from its population. The expanded population adds $1{,}000 - 800 = 200$ loans.

   (b) Each parameter describes all loans in the relevant population (800 or 1,000). Each statistic describes the 50 selected loans in its column. The two balance measures are in million COP.

Measure and paired symbols	Initial sample	Broader sample
Mean outstanding balance μ and $\bar{x}$	8.0	8.8
Standard deviation of outstanding balances σ and s	2.4	2.9
Share with an overdue installment p and $\hat{p}$	$10/50 =$ 20%	$8/50 =$ 16%

   (c) The broader sample has a higher mean balance ($8.8 - 8.0 = +0.8$ million COP), a higher standard deviation ($2.9 - 2.4 = +0.5$ million COP), and a lower overdue share ($16\% - 20\% = -4$ percentage points). The parameters describe a broader population. Whether or by how much their values differ is unknown from the samples alone.